

MASTER'S COMPREHENSIVE EXAMINATION

Part II Applied (2 hours) December 3rd, 2016.

Instructions: Attempt any three out of the following four questions on both pages.
Mention which three you would like to have graded. Open book and open notes.

1. People want to compare two plantations for pesticide levels in bananas. They have tested the levels of 6 randomly chosen bunches of bananas each from these two plantations. From each bunch, two randomly chosen bananas are tested. The following pesticide level results are observed for the two bananas within each Bunch

PLANTATION A - Bunch-1 20, 24 Bunch-2 30, 24 Bunch-3 19, 19 Bunch-4 12, 18
Bunch-5 27, 21 Bunch-6 15, 23

PLANTATION B - Bunch-1 20, 14 Bunch-2 20, 24 Bunch-3 16, 14 Bunch-4 17, 13
Bunch-5 17, 15 Bunch-6 19, 17

You can assume the observations are independently normally distributed with equal variance

- A. Give a 95% confidence interval for the mean level of pesticides in Bananas from Plantation A
- B. Give the best statistical test of whether mean pesticide levels differ in Plantation A and B at a two sided $\alpha=0.05$

2) Anchovies (a small fish) are a famous chefs delight. The chef rates the taste quality of 5 different brands of anchovies. The brands came packaged in both a jar and a can. The chef rates 10 combinations of brand/package from 1 to 10 (where 10 is the best taste).

The following is the ratings:

Brand	Rating
Agostino Recca (Jar)	2
Agostino Recca (Can)	4
Crown Prince (Jar)	1
Crown Prince (Can)	2
Merro (Jar)	6
Merro (Can)	9
Ortiz (Jar)	8
Ortiz (Can)	10
Roland (Jar)	1
Roland (Can)	3

We want to test if there is evidence that the method of packing (Jar versus Can) is related to a difference between the tastiness (to the Chef) of anchovies. State the Null Hypothesis State the Alternative Hypothesis Explain what test statistic you would use, compute it, and explain how you would get a p-value for it.

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3. A researcher at a large university hypothesizes that the proportion of students who spend more than \$200 in the past semester on books is greater for students in the social sciences than for students in the humanities.

The researcher draws independent random samples of 60 students from each of the areas:

Humanities Social Sciences

Statistic	Humanities	Social Sciences
sample size	60	60
Median	\$160	\$200
Q1	\$120	\$170
Q3	\$200	\$246

Carry out a test at the 0.05 level of significance to see if there is evidence to support the researcher's hypothesis. The Null Hypothesis is that there is no difference between humanities and social sciences in the proportion of the Universities students in the past semester who spend more than \$200 on books.

4. Three types of Fish live in Rutgers pond; Busch-fish, Livingston-fish and Douglas-fish. Drs. HD and DH each recorded fish caught in Rutgers pond from randomly chosen samples.

- Out of a random sample of 120 fish recorded by Dr. HD, 40 were Douglas, 40 were Livingston and 40 were Busch.
- Dr. DH recorded results from a different independent random sample. But he did not record the number of Busch fish Thus out of 90 fish that were either Douglas or Livingston, Dr. HD recorded 60 Livingston and 30 Douglas

Assume that the fish recorded by Drs. HD and DH both random, independent of each other and obtained under the same probabilities to be a Douglas, Livingston or a Busch.

- A. Use this data from Drs. HD and DH to find the best point estimates for the probability that the species of a fish caught at Rutgers pond were i) Douglas, ii) Livingston and iii) Busch
- B. Give a large sample 95% confidence interval for the probability that the species of a fish caught at Rutgers Pond is a Busch